WORKED SOLUTION

# STEP II 2021, Question 9

Mechanics • Statics

**STEP II 2021, Question 9**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

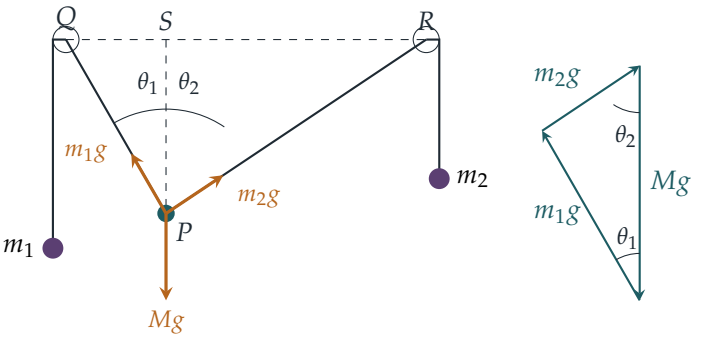
www.ocr.org.uk/Images/696588-2021-past-papers.zip

## The key idea

Because the pulleys are smooth, each string section carries the weight of the particle hanging at its far end, so the three forces on $P$ have magnitudes in the ratio $m_1 : m_2 : M$. Those three forces close up into a triangle whose sides are $m_1 g$, $m_2 g$ and $Mg$, and whose angles are built from $\theta_1$ and $\theta_2$. The cosine rule on that triangle then gives everything: the inequalities come from asking that the triangle exists with both angles acute, and the ratio comes from $\tan\theta_1$ and $\tan\theta_2$.

## Solution

**Model.** The particles are point masses, the string is light and inextensible, and the pulleys are small and smooth. So the tension in $PQ$ equals the tension in the vertical part supporting $m_1$, which is in equilibrium: $T_1 = m_1 g$. Likewise the tension in $PR$ is $T_2 = m_2 g$. Write $h = SP$.



**Part (i)**

Resolving horizontally and vertically at $P$ (and cancelling $g$):

$$m_1 \sin\theta_1 = m_2 \sin\theta_2, \qquad m_1 \cos\theta_1 + m_2 \cos\theta_2 = M. \tag{$*$}$$

Equivalently, the three forces placed nose to tail form a closed triangle with sides $m_1 g$, $m_2 g$, $Mg$ (right of the figure). The side $Mg$ is vertical, so the angle between the sides $Mg$ and $m_1 g$ is $\theta_1$ (opposite the side $m_2 g$), and the angle between $Mg$ and $m_2 g$ is $\theta_2$ (opposite $m_1 g$). The third angle, opposite $Mg$, is $180^\circ - \theta_1 - \theta_2$.

If you prefer algebra to the picture, the same facts drop out of $(*)$: write $m_1 \cos\theta_1 = M - m_2 \cos\theta_2$ and $m_1 \sin\theta_1 = m_2 \sin\theta_2$, square and add:

$$m_1^2 = M^2 + m_2^2 - 2Mm_2 \cos\theta_2, \quad \text{so} \quad \cos\theta_2 = \frac{M^2 + m_2^2 - m_1^2}{2Mm_2},$$

and, swapping the roles of the two sides,

$$\cos\theta_1 = \frac{M^2 + m_1^2 - m_2^2}{2Mm_1}.$$

**(a)** The triangle of forces is a genuine (non-flat) triangle, because $\theta_1 + \theta_2$ lies strictly between $0^\circ$ and $180^\circ$. The triangle inequality is therefore strict, and gives $M < m_1 + m_2$.

For the lower bound we use the fact that $\theta_2$ is acute, so $\cos\theta_2 > 0$. Since $2Mm_2 > 0$, the formula above gives

$$M^2 + m_2^2 - m_1^2 > 0 \iff M^2 > m_1^2 - m_2^2 \iff M > \sqrt{m_1^2 - m_2^2},$$

where the last step is valid because $M > 0$ and $m_1^2 - m_2^2 > 0$. Together,

$$\sqrt{m_1^2 - m_2^2} < M < m_1 + m_2.$$

? **Ask yourself** Why does the condition on $\theta_2$ matter, and not the one on $\theta_1$?

In a triangle the larger side faces the larger angle. Since $m_1 > m_2$, the angle $\theta_2$ (opposite $m_1 g$) is larger than $\theta_1$ (opposite $m_2 g$). So once $\theta_2$ is acute, $\theta_1$ is automatically acute too; the only real restriction is on $\theta_2$. (Indeed $\cos\theta_1 > 0$ holds for every $M > 0$, as $m_1 > m_2$.)

**(b)** In the right-angled triangles $QSP$ and $RSP$, with $SP = h$,

$$QS = h\tan\theta_1, \qquad SR = h\tan\theta_2.$$

Hence

$$r = \frac{QS}{SR} = \frac{\tan\theta_1}{\tan\theta_2} = \frac{\sin\theta_1}{\sin\theta_2}\cdot\frac{\cos\theta_2}{\cos\theta_1}$$
$$= \frac{m_2}{m_1}\cdot\frac{M^2+m_2^2-m_1^2}{2Mm_2}\cdot\frac{2Mm_1}{M^2+m_1^2-m_2^2} \quad \text{(using } (*) \text{ and the cosines)}$$
$$= \frac{M^2-m_1^2+m_2^2}{M^2-m_2^2+m_1^2},$$

which is the required ratio. Both brackets are positive by part (a), so $r > 0$, and $r < 1$ because $m_1 > m_2$: the point $S$ is nearer $Q$, as the figure suggests.

#### Part (ii)

The cosine rule for the side $Mg$ of the triangle of forces, whose opposite angle is $180^\circ - (\theta_1+\theta_2)$, reads

$$M^2 = m_1^2 + m_2^2 + 2m_1m_2\cos(\theta_1+\theta_2).$$

(Algebraically: square and add the two equations in $(*)$.) With $M^2 = m_1^2+m_2^2$ this forces $\cos(\theta_1+\theta_2) = 0$. Since $\theta_1$ and $\theta_2$ are acute, $0^\circ < \theta_1+\theta_2 < 180^\circ$, and the only possibility is

$$\theta_1+\theta_2 = 90^\circ.$$

So the string makes a right angle at $P$. (This value of $M$ does lie in the range of part (a), so the configuration exists.)

Now the cosines simplify: $\cos\theta_1 = \dfrac{2m_1^2}{2Mm_1} = \dfrac{m_1}{M}$ and $\cos\theta_2 = \dfrac{m_2}{M}$. Since $\theta_2 = 90^\circ - \theta_1$, we get $\sin\theta_1 = \cos\theta_2 = m_2/M$, so

$$\tan\theta_1 = \frac{m_2}{m_1}, \qquad \tan\theta_2 = \frac{1}{\tan\theta_1} = \frac{m_1}{m_2}.$$

Therefore

$$\frac{QR}{SP} = \frac{h\tan\theta_1 + h\tan\theta_2}{h} = \frac{m_2}{m_1}+\frac{m_1}{m_2} = \frac{m_1^2+m_2^2}{m_1m_2},$$

so $QR : SP = (m_1^2+m_2^2) : m_1m_2$.

## Checks

- With $m_1 = 5$, $m_2 = 3$, $M = 6$ (inside $4 < M < 8$), solving $(*)$ numerically gives $\theta_1 \approx 29.93^\circ$ and $\theta_2 \approx 56.25^\circ$. Then $\tan\theta_1/\tan\theta_2 \approx 0.3846 = \frac{5}{13}$, which matches $r = \frac{36-25+9}{36-9+25} = \frac{20}{52}$.
- At the ends of the range: as $M \to \sqrt{m_1^2-m_2^2}$ we get $\theta_2 \to 90^\circ$ and $r \to 0$ (the string $PR$ becomes horizontal); as $M \to m_1+m_2$ both angles tend to 0 and $P$ sinks far below $QR$.
- Part (ii) with $m_1 = 4$, $m_2 = 3$, $M = 5$: $\tan\theta_1 = \frac{3}{4}$, $\tan\theta_2 = \frac{4}{3}$, and $QR/SP = \frac{25}{12}$. Check $(*)$: $4\cdot\frac{3}{5} = 3\cdot\frac{4}{5}$ and $4\cdot\frac{4}{5}+3\cdot\frac{3}{5} = 5$.

## Common slips

- Taking the angle opposite $Mg$ in the triangle of forces to be $\theta_1 + \theta_2$ rather than $180^\circ - (\theta_1 + \theta_2)$, which flips the sign in the cosine rule.
- Using a non-strict triangle inequality, or forgetting that the lower bound needs the angles to be acute.
- Inverting the ratio: $r = QS/SR$, and $QS$ is the side next to the heavier particle.

## Beyond STEP

The triangle of forces is the simplest case of a general fact: forces in equilibrium sum to the zero vector, so drawn nose to tail they form a closed polygon. Engineers use this idea (the "force polygon") to analyse frameworks, and in first-year mechanics it reappears as the vector form of statics, alongside the condition that moments also balance. For more, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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